

Language Model Embeddings for Species Synonym Detection in Biodiversity

Sefika Efeoglu^{1,2,*}, Jana Hoffmann¹

¹Senckenberg – Leibniz Institution for Biodiversity and Earth System Research, 15374 Müncheberg, Germany

²Institute of Informatics, Freie Universität Berlin, 14195 Berlin, Germany

Abstract

Scientific species names often become synonymous because of taxonomic revisions, independent descriptions, and changes in biological classification. Such ambiguity hinders the integration of biodiversity data across heterogeneous resources and affects the quality of biodiversity knowledge bases. This work investigates using language model embeddings for zero-shot species synonym detection. We harvest botanical and entomological species data from the Global Biodiversity Information Facility and construct a dataset covering six taxonomic orders. We benchmark 21 embedding models against a Levenshtein-distance baseline, using cosine similarity for zero-shot binary classification. Our evaluation shows that the strongest-performing models include retrieval-oriented embeddings, which outperform lexical string matching and several domain-specific biomedical language models. Among the evaluated models, *BGE Base* (English version 1.5) achieves the best overall performance across all six taxonomic orders. The results indicate that general-purpose retrieval embeddings capture semantic similarity between taxonomic entities more effectively than models specialized for biomedical text, making them a promising approach for large-scale biodiversity knowledge integration and taxonomy curation¹.

Keywords

Species Matching, Species Synonymy, GBIF, Biodiversity

1. Introduction

Ambiguity in species names is a common challenge in biodiversity informatics. The same species may be referred to by multiple scientific names due to taxonomic revisions, independent descriptions, or changes in classification. These synonyms complicate the integration of biodiversity data from heterogeneous sources and affect the quality of biodiversity knowledge resources [1]. Furthermore, ambiguity may arise from changes in taxonomic rank, e.g., class, order, genus, subgenus, or species. Cross-domain ambiguity also arises when identical scientific names occur in different organismal groups and across different taxonomic ranks. For instance, *Victoria* refers to a plant genus that includes the giant water lilies, while *Victoria* is also the name of an insect genus. Therefore, accurate synonym matching is essential for integrating biodiversity data and maintaining taxonomic knowledge bases, i.e., the Global Biodiversity Information Facility (GBIF)¹. Another potential application is identifying species synonyms within text corpora and entity linking across different knowledge graphs.

Language models have demonstrated strong performance on ontology matching [2, 3]. More recently, Retrieval-Augmented Generation (RAG) has also been applied to these problems [4, 5]. However, these approaches [2, 4, 3, 5] benefit from the parametric memory of large language models (LLMs) and require substantial computational resources or incur API usage costs. Consequently, their adoption may impose additional financial burdens on life scientists. On the other hand, several embedding models have been specifically designed for semantic retrieval, e.g., BAAI General Embeddings (BGE)², making

¹The repository: https://github.com/Senckenberg-Nature-Research-SDEI/lm_embeddings4species

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*Corresponding author.

✉ sefika.efioglu@senckenberg.de (S. Efeoglu); jana.hoffmann@senckenberg.de (J. Hoffmann)

ORCID 0000-0002-9232-4840 (S. Efeoglu); 0000-0001-6214-661X (J. Hoffmann)



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¹<https://www.gbif.org/>, accessed on July 13, 2026.

²<https://bge-model.com/>, accessed on 20.07.2026

them promising candidates for species synonym detection while requiring only modest computational resources, thereby improving accessibility for life scientists.

In this paper, we benchmark embeddings from 21 language models, including domain-specific models (e.g., SciBERT), for zero-shot species synonym detection using data harvested from GBIF. We construct a dataset spanning six orders (taxonomic rank) across the botanical and entomological (insect) domains, since both have more common species names. These domains also share homonyms, meaning identical scientific names can refer to different organisms. Then, we compare the models against a Levenshtein baseline. Negative examples are generated by randomly sampling pairs of non-synonymous species. We focus on the zero-shot setting because training or fine-tuning a custom model requires expertise many life scientists may not have. Our main finding reveals that **BGE Base (English v1.5)**³ achieves the best overall performance.

The remainder of this paper is organized as follows. Section 2 reviews related work on ontology and species name matching. Section 3 describes the data preparation process and the proposed zero-shot species synonym detection. Section 4 presents the experimental settings. Section 5 reports and discusses the experimental results across the six orders (taxonomic rank). Finally, Section 6 concludes the paper and outlines directions for future work.

2. Related Work

In this section, we discuss related work on ontology matching and synonym detection, specifically species name matching, in the biodiversity domain.

For ontology matching, previous studies [6, 7, 8, 9] have proposed attention-based approaches and Siamese architectures utilizing language model embeddings like BERT. These approaches generate contextual representations of the concepts or entities and achieve remarkable performance on ontology matching. Additionally, with the rise of large language models (LLMs) like GPT and Llama 2, [2, 4, 10] examine LLM-based approaches, including zero-shot prompting and RAG-based prompting, utilizing contextual information generated from subgraphs of the source and target entities.

In addition to general ontology matching approaches, we also summarize the matching approaches that attempt to resolve the (synonym) matching problem in the biodiversity domain. The Ontology Alignment Evaluation Initiative (OAEI) Biodiversity and Ecology track addresses the matching of large domain-specific ontologies [1], e.g., SWEET and ENVO, and taxonomic rank-based matching has been proposed to address scalability issues [11]. However, existing OAEI biodiversity benchmarks do not consider biodiversity knowledge resources such as GBIF, Catalog of Life (COL)⁴, and Barcode of Life⁵. Life scientists, especially taxonomists, commonly use these resources. In contrast, this work evaluates species matching using data from a real-world biodiversity resource.

In this work, we benchmark language model embeddings with zero-shot binary classification on species names at six *order* (taxonomic rank) levels in entomology and botany. We aim to identify the best-performing language model embeddings on domain-specific data to inform future research directions, such as entity linking.

3. Methodology

This section describes how species data are harvested and evaluated using language model embeddings for zero-shot binary classification. Figure 1 provides an overview of the proposed workflow, from data harvesting to detecting species synonyms. The workflow takes pairs of scientific species names from either the botanical or entomological domain as input and predicts whether they are synonyms or non-synonyms. We first describe the data preparation process and how we construct the binary

³<https://huggingface.co/BAAI/bge-base-en-v1.5>, accessed on July 13, 2026.

⁴<https://www.catalogueoflife.org/>, accessed on 23.07.2026.

⁵<https://www.boldsystems.org/>, accessed on 23.07.2026.

classification dataset. We then present a zero-shot classification approach that generates embeddings for species names and classifies each pair using cosine similarity and a fixed similarity threshold.

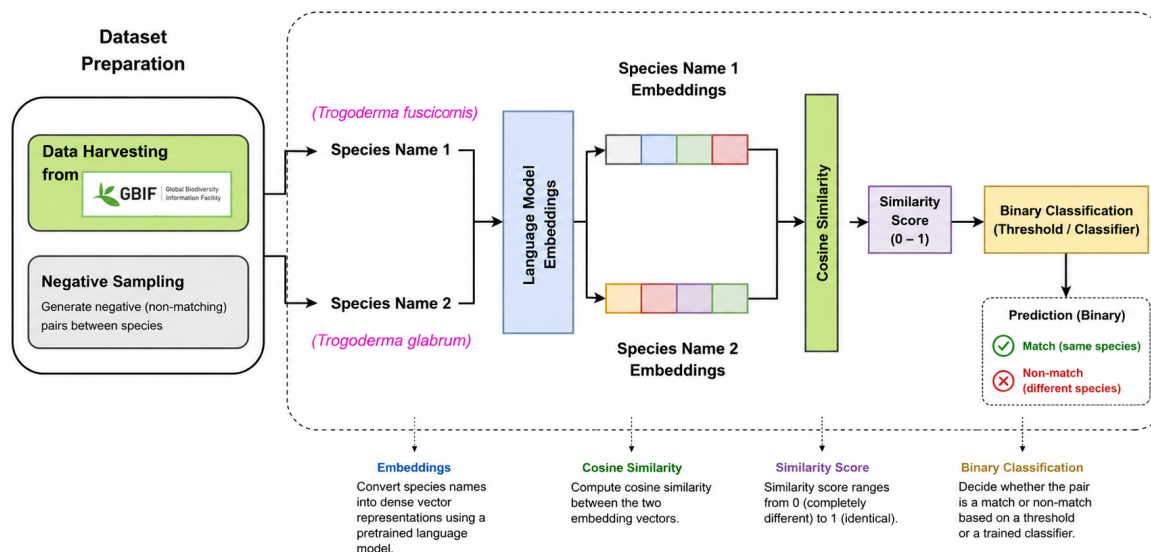


Figure 1: Overview of the proposed workflow for zero-shot species synonym detection. Note: OpenAI created the figure based on the original figure.

3.1. Data Preparation

The Global Biodiversity Information Facility (GBIF) is a widely used biodiversity data infrastructure supported by governments worldwide. We harvest botanical and entomological data using the GBIF Python API⁶. Specifically, we select two *orders* (taxonomic rank) from the entomological domain and four *orders* from the botanical domain. They represent diverse *orders* with different synonym distributions and varying dataset sizes.

For each *order*, we harvest accepted species names together with their recorded synonyms from GBIF. Positive samples consist of accepted name-synonym pairs, whereas negative samples are generated by randomly pairing non-synonymous species names, which are accepted names by GBIF. To obtain a balanced binary classification dataset, we sample an equal number of synonym and non-synonym pairs per *order*. Note that we consider only species names, not biological classification, when computing embeddings for our zero-shot binary classification.

3.2. Zero-Shot Binary Classification

We evaluate language model embeddings by generating vector representations for species names. For each pair of species names, we compute their embeddings and measure their semantic similarity using *cosine similarity*. We compare the resulting similarity score against a fixed threshold to classify the pair as either a synonym or a non-synonym. This approach enables zero-shot species synonym detection without requiring task-specific model training or fine-tuning.

4. Experimental Settings

This section first presents the data-harvesting settings and then the zero-shot classification settings.

⁶<https://www.gbif.org/tool/OlyoYyRbKCScMKIi4oIT/pygbif-gbif-python-client>, accessed on July 17, 2026.

4.1. Harvesting Settings

Species records are harvested in batches of up to 2,000 entries per API request until a maximum of 100,000 records is reached for each *order* (taxonomic rank). To avoid overrepresenting highly speciose genera, we randomly sampled up to 100 species from each genus with more than 100 species, using a seed of 42 (see details in Table 1). This sampling strategy yields a large, diverse collection of species while maintaining balanced representation across genera. An overview of the resulting dataset [12] is in Table 2, which is harvested according to the given key in Table 1. Lastly, GBIF data are published under three licenses: CC0, CC BY, and CC BY-NC.

Table 1

GBIF root keys used to harvest each *order*.

Order (<i>Taxonomic Rank</i>)	Lepidoptera	Coleoptera	Rosales	Fabales	Lamiales	Poales
GBIF root key	797	1470	691	1370	724	1369

Table 2

Overview of the datasets used for synonym classification. The sample counts include balanced synonym and non-synonym species pairs. We harvested a subset of these datasets in [13, 14, 15, 16, 17, 18]

Domain	Order	Number of samples
Botany	Fabales [13]	11,388
	Lamiales [14]	492
	Poales [15]	26,264
	Rosales [16]	18,836
Entomology	Coleoptera [17]	546
	Lepidoptera [18]	2,948
Total		60,474

4.2. Zero-Shot Classification Settings

We evaluate 21 language model embeddings, listed in Table 4 in the Appendix section, against a Levenshtein baseline, which measures string similarity based on the minimum number of single-character edits required to transform one string into another. We selected models to cover different domains and parameter sizes. For binary classification, we use a fixed cosine similarity threshold of 0.70 across all models and datasets, including the Levenshtein baseline, without task-specific threshold optimization. In addition to the binary predictions, we retain the cosine similarity scores for all species pairs and analyze their distributions to provide a more comprehensive evaluation of the embedding models.

5. Results and Discussion

We evaluate 21 models (see Table 4) against the Levenshtein baseline on six *orders* (taxonomic ranks) spanning the botanical and entomological domains. Levenshtein distance serves as the baseline for lexical string similarity. Table 3 reports the F1 scores achieved by each model across *orders*.

Among all evaluated models, *bge-base-en-v1.5*⁷ consistently achieves the highest F1 score across all six *orders*, demonstrating robust performance for zero-shot species synonym detection. In contrast, the Levenshtein baseline produces substantially lower F1 scores than the best-performing embedding models, highlighting the advantage of semantic embeddings over lexical string similarity. Although

⁷<https://bge-model.com/index.html>, accessed on 23.07.2026.

lexical similarity can capture minor spelling variations, it may be insufficient for synonym pairs that differ substantially at the string level.

Retrieval-oriented embedding models generally outperform domain-specific biomedical models, including ModernBERT-bio-large, BiomedBERT, S-BioBERT, and SciBERT (both cased and uncased). Interestingly, models specifically pretrained on biomedical corpora do not consistently provide better synonym detection performance despite the biological nature of species names. PatentSBERTa is the second-best-performing model across most *orders*, suggesting that models trained on technical terminology can also effectively capture semantic relationships among scientific names. The lightweight *all-MiniLM-L6-v2* model achieves lower performance than most language model embeddings and performs below the Levenshtein baseline for some *orders*. Additionally, several models obtain F1 scores close to 0.667, which is consistent with classifying most pairs as synonyms under the fixed threshold on the balanced datasets.

Table 3

F1 score by model and taxonomic order. The best score for each order is shown in bold, and the baseline results are underlined. More information about the models is in Table 4.

Model	Entomology		Botany				Average
	Coleoptera	Lepidoptera	Fabales	Lamiales	Poales	Rosales	
Levenshtein (Baseline)	<u>0.4862</u>	<u>0.2850</u>	<u>0.2422</u>	<u>0.2837</u>	<u>0.2437</u>	<u>0.2453</u>	<u>0.2977</u>
ModernBERT-bio-large	0.6667	0.6667	0.6667	0.6676	0.6667	0.6667	0.6669
BiomedBERT	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667
gpt2-medium	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667
scibert_scivocab_uncased	0.6756	0.6853	0.6796	0.6781	0.6765	0.6714	0.6777
bge-base-en-v1.5	0.8186	0.8081	0.7817	0.9456	0.7506	0.7362	0.8068
all-mpnet-base-v2	0.6203	0.5143	0.4806	0.2316	0.4534	0.5557	0.4760
bge-reranker-large	0.6650	0.6671	0.6661	0.6685	0.6664	0.6654	0.6664
bge-reranker-v2-m3	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667
PatentSBERTa	0.7682	0.7458	0.6762	0.7190	0.6597	0.7254	0.7157
bge-reranker-base	0.6675	0.6670	0.6678	0.6628	0.6646	0.6627	0.6654
S-BioBERT	0.4563	0.3520	0.3646	0.4673	0.3169	0.3172	0.3790
bge-large-zh-v1.5	0.6618	0.6508	0.5674	0.5000	0.5417	0.5947	0.5861
sciNCL	0.6699	0.6719	0.6697	0.6676	0.6686	0.6684	0.6694
scibert_scivocab_cased	0.7043	0.6995	0.6871	0.6866	0.6821	0.6816	0.6902
bge-reranker-v2-gemma	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667
ModernBERT-large	0.6634	0.6670	0.6669	0.6649	0.6667	0.6675	0.6661
bge-large-en	0.6683	0.6680	0.6669	0.6676	0.6674	0.6671	0.6675
all-MiniLM-L6-v2	0.4807	0.3922	0.3899	0.2222	0.3853	0.4092	0.3799
e5-base-v2	0.6667	0.6667	0.6668	0.6667	0.6668	0.6667	0.6667
gpt2	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667	0.6667
llm-embedder	0.6725	0.6700	0.6695	0.6703	0.6716	0.6700	0.6707

Overall, the strongest-performing models include retrieval-oriented embeddings, with BGE Base achieving the highest performance across all six taxonomic orders.

6. Conclusion

This paper benchmarked language model embeddings for zero-shot species synonym detection in biodiversity. Using data harvested from GBIF, we constructed balanced datasets spanning six taxonomic orders across the botanical and entomological domains and evaluated 21 embedding models against a Levenshtein-distance baseline. Our experimental results show that the strongest-performing models include retrieval-oriented embeddings, which outperform lexical string matching and several domain-specific biomedical models. Among the evaluated models, BGE Base (English v1.5) achieves the best overall performance across all six taxonomic orders, demonstrating that general-purpose retrieval embeddings effectively capture semantic relationships between scientific species names. Moreover, the proposed approach requires neither task-specific fine-tuning nor domain-specific training, making it a practical solution for large-scale biodiversity knowledge integration and taxonomy curation.

Despite these promising results, this work has several limitations that future research will address. In zoological nomenclature, including that used in entomology, scientific names are typically represented as a combination of the species name, author name, and year of publication. This differs in botanical nomenclature, where scientific names combine only the species and author names. The latter two components are particularly important for cross-domain synonym detection because identical scientific names may refer to different taxa described by different authors. To address this challenge, we plan to extend our approach with graph-based contextual embeddings that incorporate taxonomic context and improve generalization across biodiversity domains. To improve model performance, we aim to fine-tune the best-performing BGE models on cross-domain species data. In addition, our current evaluation is limited to two domains. As future work, we intend to investigate additional biodiversity resources, perform cross-domain entity matching, explore vernacular (trivial) names, e.g., robin for “*Erithacus rubecula*” (Latin binomial name version), and study adaptive similarity threshold selection. We also aim to adapt our methodology for similar nomenclature-related domains beyond biodiversity research, e.g., mineralogy and meteoritics.

Declaration of Generative AI Use

During the preparation of this work, a generative AI model (GPT 5.5 with a temporary chat option) was used to check grammar and spelling. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the publication’s content. Figure 1 has been modified by OpenAI based on our original figure.

Acknowledgement.

We used the central High Performance Computer system at Freie Universität Berlin to conduct experiments in the botanical domain, and we thank them for the resources provided.

Appendix

Table 4

Embedding and reranking models evaluated for taxonomy synonym detection.

Model	HuggingFace Model Card	Category	Parameter Size
bge-base-en-v1.5	https://huggingface.co/BAAI/bge-base-en-v1.5	General-purpose	109M
PatentSBERTa	https://huggingface.co/AI-Growth-Lab/PatentSBERTa	Technical	110M
scibert_scivocab_cased	https://huggingface.co/allenai/scibert_scivocab_cased	Scientific	110M
scibert_scivocab_uncased	https://huggingface.co/allenai/scibert_scivocab_uncased	Scientific	110M
all-mpnet-base-v2	https://huggingface.co/sentence-transformers/all-mpnet-base-v2	General-purpose	110M
sciNCL	https://huggingface.co/malteos/scincl	Scientific	110M
llm-embedder	https://huggingface.co/BAAI/llm-embedder	Retrieval	109M
bge-large-zh-v1.5	https://huggingface.co/BAAI/bge-large-zh-v1.5	Multilingual	326M
all-MiniLM-L6-v2	https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2	Lightweight	22M
S-BioBERT-snli-multinli-stsb	https://huggingface.co/pritamdeka/S-BioBert-snli-multinli-stsb	Biomedical	108M
BiomedNLP-BiomedBERT-base-uncased-abstract	https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract	Biomedical	110M
bge-large-en	https://huggingface.co/BAAI/bge-large-en	General-purpose	335M
e5-base-v2	https://huggingface.co/intfloat/e5-base-v2	Retrieval	109M
ModernBERT-bio-large	https://huggingface.co/almanach/ModernBERT-bio-large	Biomedical	396M
ModernBERT-large	https://huggingface.co/answerdotai/ModernBERT-large	General-purpose	395M
gpt2-medium	https://huggingface.co/openai-community/gpt2-medium	Decoder-only	355M
gpt2	https://huggingface.co/openai-community/gpt2	Decoder-only	124M
bge-reranker-large	https://huggingface.co/BAAI/bge-reranker-large	Reranker	560M
bge-reranker-base	https://huggingface.co/BAAI/bge-reranker-base	Reranker	278M
bge-reranker-v2-m3	https://huggingface.co/BAAI/bge-reranker-v2-m3	Reranker	568M
bge-reranker-v2-gemma	https://huggingface.co/BAAI/bge-reranker-v2-gemma	Reranker	2.51B

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